

Swaroop Thikane

Abdul Lat, Maharashtra | 9766560792 | swaroopthikane@gmail.com | [LinkedIn](#) | [Portfolio](#)

Education

B.Tech — Electronics & Telecommunication Engineering

Expected 2026

D.K.T.E's Textile & Engineering Institute, Ichalkaranji | CGPA: 7.82

Technical Skills

Languages: C, C++, Embedded C, Java, Verilog, VHDL, Python
MCU Platforms: ESP32, ESP32-S2, STM32, Raspberry Pi CM4, PIC12LF1572, ATmega (164PU, 3209)
Protocols: UART, SPI, I2C/TWI, GSM, GPS, RFID, BLE, Cellular IoT, USB Host (CDC-ECM)
Embedded Topics: Interrupt-driven FSMs, Timer peripherals, Low-power design, Watchdog, Real-time constraints
Tools: PCB Design, Oscilloscope, Logic Analyzer, MPLAB X, ESP-IDF, Atmel Studio, Proteus ISIS, Git

Projects

SmartBag

- Developed a smart bag prototype with location tracking and inventory management using GSM, GPS, and RFID modules; mastered GSM/GPS module interfacing and RFID-based identification systems
- Utility Patent Published:** Smart Bag System for Enhanced Safety, Organization and Educational Engagement (Application No. 202421041453)

Filament Maker

- Designed a system to convert waste PET bottles into 3D-printable filaments, implementing stepper motor control, motor drivers, and relay-based switching for extrusion temperature and feed regulation

IoT Based Pearl Farm Health Regulator

- Built a real-time monitoring and regulation system for pH, TDS, and dissolved oxygen levels in pearl farm water; automated green algae feeding actuation critical for pearl formation
- Awarded TIH-IoT Chanakya Fellowship 2024 (**Rs. 6 Lakhs**); **Design Patent Granted** (No. 454051-001); **Research paper published** on IEEE Xplore Digital Library

KrishiCobot

- Agriculture-assistive robot using a UR5 robotic arm and camera integration for automated fertilizer refilling, crop shape detection, and fruit sorting (good vs. rotten)
- Implemented embedded control logic for real-time plant condition detection and actuator triggering based on sensor and vision inputs

Internship Experience

Embedded Systems Intern — SCI COM Software Pvt. Ltd.

2025-26

- Developed firmware targeting STM32 and ESP32 for a DRDO high-temperature precise timing system; worked extensively with I2C, SPI, and UART peripheral configuration
- Designed and validated PCBs; performed hardware bring-up and debugging using oscilloscope and logic analyzer; integrated sensors and power management sub-circuits

Industrial Training — Kulkarni Power Tools Ltd.

2024

- Studied IGBT-based welding machine architectures, power electronics schematics, and mains-to-DC conversion circuits (240V AC to rectified DC); analysed opto-coupler power supply designs

Certifications, Achievements & Leadership

- Certifications: Introduction to IoT — IIT Bombay | Principles of Management — NPTEL | CLA Programming Essentials in C
- Achievements: Mini Project Competition 2023-24 Winner | DKTE Ideathon Winner | NRIT Ideathon Finalist | Best Idea Presenter at Ideation Workshop
- Leadership: IEEE SB Chapter Treasurer — coordinated events, managed funding, collected sponsorships | COMSA President — organised technical workshops, competitions, and student activities